

**PUBPOL 4390/5390
PROBLEM SET 2****Due 11:55pm, Monday November 3, via Canvas**

- Work in teams of 1-3. You can choose your own teams. Clearly indicate all the members of the team at the beginning of the problem set.
- Only one submission to Canvas per team, please.
- Turn in the main problem set as one file only, not several documents. PDF preferred. Clearly label the problem set file (example: “PS 1 – PADM 5390 – Olivero Miller.pdf”).
- Please also upload, as auxiliary files, the programs used in your computations (eg Stata’s .do files).

1. Predicting Birthweight - continued

Use dataset “**Gsmall**” to train OLS models predicting birthweight, and when you are satisfied, use dataset H as an out-of-sample dataset to assess your models’ performance. Important – don’t use the results in dataset H to choose among candidate models; instead just use dataset H once at the end, to measure goodness of fit, after you have committed to your models.

I would like you to use 3 different sets of covariates, based on the following principles:

Model #		# of covariates	
1	Small	$K \leq 10$	Keep it simple, but still try to get good predictions.
2	Moderate	$20 < K < 50$	
3	Large	$200 < K$	Include polynomial or spline transformations, rich interactions between variables, etc. Try to build a model rich enough to have OLS do some over-fitting, even with this larger dataset!

Note: the data are different than in Problem Set 1. I have done some pre-cleaning here. The covariates might be different than the ones you used for PS 1.

- You can update your covariate sets for your 3 main models from PS 1, based on what is available.
- For each of the covariate sets, estimate the following prediction models, training on Gsmall.
 - OLS
 - LASSO
 - Ridge
 - Elastic Net
- You can use the default Cross-validation process for choosing the tuning parameters. Learn what those defaults are, and briefly describe the cross-validation procedure. (e.g. how many folds? What is the measure of goodness-of-fit that is used to select a model? What else would a researcher need to know to reproduce this process, potentially using different software?)

- Compute and show RMSE in the out-of-sample (Dataset H) for each of the covariate sets, for each of the prediction models. For each of the ML models, also report in the table on the value of the tuning parameters that you use.
- How do each of the ML models do, compared to OLS? Does this depend on which covariate set you are using? What is the best model, comparing across all ML choices and covariate sets?

2. Predicting House Prices - continued

This problem builds on the data used in the JEP article by Mullainathan and Spiess, <https://www.aeaweb.org/articles?id=10.1257/jep.31.2.87> . The replication package link is at: <https://www.openicpsr.org/openicpsr/project/113993/version/V1/view>

The replication package appears to be “code only”, and I don’t see the data that they use. In their online appendix (<https://www.aeaweb.org/content/file?id=4212>), section B, they describe how they create their sample. You can use a similar process – you will have a different sample partition of “train” and “hold out” than they do. When you randomly sort the data to split into “train” and “hold out”, make sure to set a random number seed first, so that you can replicate your results.

I think that the microdata for the 2011 AHS can be downloaded from here: <https://www.census.gov/programs-surveys/ahs/data/2011/ahs-2011-public-use-file-puf/ahs-2011-national-public-use-file-puf.html> . I *think* (but I’m not yet 100% sure) that we want to work with the PUF National Flat File.

- Depending on how your PS 1 replication went, you can re-visit your data cleaning steps. See if you can get a close sample size and basic OLS results that match pretty closely to what is reported in the paper.
- In PS 1, we examined both R^2 and RMSE. We’ll continue to look at both in this exercise. We did this for both in-sample and out-of-sample. We’ll continue this, as well. In PS 1, you created a “replication” model and a “kitchen sink” model. We’ll continue to use these two models. You can update/change them if you want. Re-report on the contents of these models – what does a reader need to know about the features/covariates?
- For both covariate sets, estimate a LASSO prediction model. Build out results in parallel to the “LASSO” row in Table 1 of the published paper. (Again – you don’t need to build a confidence interval for the R^2).
- Build out the “percent improvement” over (replication OLS) results, by quintile, for each of the LASSO models. For each model, use (OLS with “replication covariates”) as the baseline to compare any improvement to. Discuss your findings.
- Repeat steps (c)-(d), for an Elastic Net model. Again, do this for both “replication covariates” and “kitchen sink covariates”.

3 – Binary LHS outcome

Return to the birth data, using **Dataset G (not Gsmall)** for training and Dataset H for testing. Create a 0/1 “Very Low Birthweight” measure that = 1 when (birthweight < 1500 grams). (Note, the inequality is strict, and 1500 grams exactly should be coded as a 0 for this variable.)

- a. We’ll start with a Probit model, using your “small” covariate set from problem 1.
- b. For the testing data (H), produce and interpret in words (to the degree possible) the following model fit assessments:
 - Calibration Plot
 - Confusion matrix
 - Accuracy Rate
 - True Positive Rate
 - Precision
 - ROC Curve
 - AUC
- c. Now train a LASSO Probit model, using your “large” covariate set, with training data Dataset G. Then produce and interpret in words (to the degree possible) the same set of model fit assessments as in part (b). For each measure, compare how the “large covariates” LASSO Probit model does compared to the “small covariates” basic Probit.